

Introduction

Background Problem

Crew is an airline’s second-largest operating cost after fuel, so how flights are staffed has an outsized effect on profitability. The cabin crew pairing problem asks for a set of legal duty sequences – *pairings* – that each begin and end at a crew member’s home base, respect working-time regulations, and together cover every scheduled flight at minimum cost. What makes cabin crew harder than cockpit crew is heterogeneity: cabin crew are cross-qualified across aircraft types and are split into classes, and the number of each class a flight needs varies with aircraft size and cabin layout. A regional turboprop may need a single attendant while a widebody needs eight.

We adopt this heterogeneity in a deliberately simple form. Every flight has a total crew requirement of between one and eight, of which exactly one must be a *senior* crew member and the rest may be anyone, normal or senior. Substitution is one-directional: a senior may take a normal seat, but a normal can never fill the senior seat. A flight that cannot be given a senior is therefore cancelled – it has no qualified crew lead on board – whereas a flight short only on normal seats is merely understaffed. This asymmetry, rather than the raw crew count, is what shapes the resourcing problem: the senior class is the scarce, binding resource.

A pairing is legal only if it obeys the operational clocks the model must track: a minimum turnaround between consecutive flights, a maximum on-duty span before a rest is required, a minimum overnight rest, and a mandatory long home break that bounds how long a crew member may stay away from base. Unlike formulations that plan around a single base and import “extra” crew at a flat penalty, we model the real geography: crew are based at many airports, can only originate and terminate duties at their own base, and must be physically positioned – by flying or deadheading – to wherever a flight departs. Coverage is thus limited not just by how many crew exist, but by whether the right class can legally reach the right place at the right time, which is the central tension the rest of this report addresses.

Original Paper’s methodology

Problem Setups

Flight Data Source

The flight schedule is real U.S. domestic data from the Bureau of Transportation Statistics (BTS) on-time performance dataset, which lists for every flight f its operating carrier, tail number, origin and destination, scheduled departure and arrival times, and distance. What BTS does not record is how many cabin crew a flight needs, so we derive it from the aircraft itself. Each tail number is joined against the FAA Aircraft Registry to recover the aircraft model and its seat count s_f – the per-tail registered value where available, otherwise a type-level estimate from FAA Type Certificate Data Sheets – and s_f is mapped to a minimum cabin crew r_f under U.S. regulation 14 CFR 121.391, where one attendant is required for every 50 seats:

$$r_f = \lceil s_f/50 \rceil$$

After enrichment the dataset spans 21 carriers, from mainline operators down to small regionals, giving a natural range of network sizes and crew requirements to test against. We treat each carrier independently: it has its own flights, airports, and crew pool, so an airline is one self-contained instance.

Code	Flights	Code	Flights	Code	Flights
WN	105,307	MQ	21,888	F9	15,526
DL	76,306	OH	21,092	PT	10,621
AA	75,088	9E	18,279	G4	9,345
OO	65,026	AS	18,163	QX	7,754
UA	62,007	B6	17,918	HA	6,690
YX	27,854	NK	17,544	YV	6,628
ZW	3,790	G7	5,575	C5	6,612

Table 1: Operating carriers in the enriched dataset and their flight counts.

A planning period of 30 days is used, extended by a return tail to a 34-day horizon so that crew committed late in the period can still be routed home; only flights departing within the planning period must be covered, the remainder sitting in the tail. We therefore take ZW as our single-crew baseline and the larger regionals and mainlines as the multi-crew cases.

Two Layer structure

Senior Layer (one senior per flight)

Normal Fill Layer ($\text{min_crew} - 1$)

Cancellation of Senior-less Flights

Senior Substitution in Idle Gaps

Per-Layer Crew Sizing

Crew Base Allocation

Senior vs Normal Crew

Formulations

Set

Data

Rolling Horizon

Seaming

Window Carry-over

Crew-Flow Model

Break-Clock Expansion

Clock-Group Aggregation

Integer Group Flow

Recovering Schedules by Flow Decomposition

Acyclic Graph gives a Clean Path Split

Coverage is Preserved

Break Requirements are Preserved

Connectivity from Flow Conservation

Interchangeability and Arbitrary Assignment

Object Function

Uncovered-Slot Penalty

Constraints

Coverage

Turnaround (45 min)

Duty Limit (14h block time since last rest)

Overnight Rest (8h)

Home Break (48h) and Away Cap (4 days)

Solve Method

Barrier vs Simplex

Results

Coverage Breakdown

Fully Crewed

Cancelled (No Senior)

Understaffed (Normal Shortfall)

Uncovered flights

Structural Spokes (isolated, sub-45 turnaround)

One-Directional Spoke Connectivity

Rolling-Horizon Seam Effects

Senior as the Binding Resource

Senior Substitution Outcomes

Two-Layer Tractability

Multi-Crew as Two Unit-Demand Layers

Effect of Barrier on Large Windows

Schedule Validation

Rule-Compliance Checks

Window-Boundary Carry-over Fix

Model Scaling Limits

Things Attempted but Left Out

Per-crew Binary

Direct Multi-Crew ($\text{min_crew} > 1$) Solve

Barrier with Crossover Disabled

Looser Time Limit

Mainline-Scale Airlines

DDD dead loop

A Separate counter for working days for each time away window

- It's basically the same thing as timeaway but blows up the model quite a bit
- merged into mandatory 48h home break every 4 days

Longer Away Window (7-day → 4-day)

- the return tail T_{tail} equals D_{AWAY} , so a 7-day away cap forces a 7-day tail → 14-day window horizon (vs 11-day at 4 days) → much larger graph/model per window
- dropped to a 4-day away cap (mandatory 48h home break), shrinking each window
- also let the separate consecutive-duty-days counter fold in (above)

Cutting Planes

vry slow, and it kinda didnt work.... got clean violations

RULE-VIOLATION SUMMARY

continuity : 0 violation(s) overlap : 0 violation(s) turnaround : 0 violation(s) d_{away} : 0 violation(s)
home_break : 0 violation(s)

TOTAL: 0 violations

but its cuz the d_{away} cut holding everyone under 4 days, the “48h break after a 4-day rotation” rule will essentially never fire — crew instead rest at home in shorter, more frequent stretches 301ed51

Warm Starts

Warning: Completing partial solution with 773441 unfixed non-continuous variables out of 773446

hint prior-window routes via `VarHintVal` instead of partial `Var.Start`, dr... ...opping Gurobi’s 80s completion sub-MIP

diff warm start attempts through three stages. F

the Gurobi log showed the original `_apply_warm_start` was setting `Var.Start=1` on only matched flight arcs and leaving every wait arc undefined, producing a near-empty partial start — Completing partial solution with 773441 unfixed ... out of 773446 (only 5 variables fixed). Gurobi reacted by running a completion sub-MIP that burned the whole time budget (0 nodes explored in subMIP for 300s, first incumbent at 247s). So I rewrote it to reconstruct each crew’s complete in-window flow by walking the wait-arc spine (depot → wait → leg → wait → leg ...) and added instrumentation reporting match rate and coverage.

That instrumentation produced the key evidence: matching was **perfect** (legs 11/11 matched, 14/14 matched) but only 11–14 legs carried across each window boundary, hinting just 0.1–0.4% of variables (2320/619830, 939/788706). The cause is structural, not a bug — each window only flies its committed 3-day region and leaves the 4-day overlap idle because tail flights aren’t in F_{cov} and so carry no coverage reward, leaving almost nothing to seed. Worse, even that sparse start still triggered the completion sub-MIP (Completing partial solution with 617795 unfixed, then 80s of 0 nodes yielding a throwaway $2.02e8$ incumbent), making the warm start net-negative: window 1 with it ran 129.6s versus window 0 without it at 106.1s. The fix was to switch from `Var.Start` to `VarHintVal` — hints feed the same guidance to Gurobi’s heuristics without invoking the completion sub-MIP, removing the 80s overhead.

Future Work / Proposals

Two-Phase (Lexicographic) Objective

Joint Senior + Normal Two-Commodity Flow

Flow-Based Senior Substitution

Coordinated Two-Layer Positioning

Scaling to Mainline-Size Airlines

Shorter Window Horizon

Coarser Time Bucketing

Column Generation

Deterministic Tie-Break for Trajectory Stability

Realistic min_crew Data